

Proposta de Projeto em Engenharia de Computadores e Informática (PECI)

Speech and Language Therapy at Home

Keywords: eHealth, Speech and Language, Communication, Virtual Meetings, Assessment, Rehabilitation, Speech Therapy, Smart Homes, Active ageing.

Supervisors: Nuno Almeida (nunoalmeida@ua.pt), António Teixeira (ajst@ua.pt)

Collaboration: Ana Rocha, Samuel Silva

CONTEXT

Ensuring access to essential healthcare services without financial hardship, and the promotion of well-being for all, in all ages, is one of the core goals established by the United Nations for 2030. Regular, proactive health monitoring is crucial for prevention and early diagnosis, empowering individuals—especially in an aging society—to manage their well-being, reduce the burden on healthcare facilities, and live independently for longer. However, current monitoring technologies are often fragmented, cumbersome, and fail to integrate seamlessly into daily life, which disrupts routines and discourages the consistent use necessary to reap their benefits. This gap highlights the need for a new approach that embeds health monitoring naturally into the home environment.

To address this challenge, the VITALITY project proposes to advance the state-of-the-art by developing a user-centric smart-home system for health monitoring with advanced interaction capabilities, such as voice and gesture controls. The goal is to create a sophisticated Information and Communication Technology (ICT) infrastructure that unifies various devices into a single, intuitive ecosystem, making health monitoring effortless and encouraging long-term adherence. Key objectives include establishing a laboratory for testing, enhancing assistants with robust security and privacy features, and progressing monitoring technologies to ultimately deliver a demonstration that showcases a more personalized, proactive, and integrated model for healthcare at home.

OBJECTIVES

Development of a platform for the smart home in construction **capable of collecting and analyzing users' information** (mainly speech) to support **assessment of communicative capabilities** (expression and comprehension) and virtual speech and hearing therapy sessions. The platform should be capable of performing continued analysis and assessment of spoken communications (speech and language) as well as supporting remote speech therapy sessions. Additionally, the platform should provide tools for **linguistic and cognitive stimulation** between remote therapy sessions, presenting to the user various linguistic and cognitive stimulation tasks.

TENTIVE WORK PLAN

- Acquisition of knowledge regarding the state-of-the-art.
- Propose and develop of Architecture and initial specification of the main modules.
- Selection of tool(s) for the development
- Iterative and user-centred development of proof-of-concept and prototypes.

Proposta de Projeto em Engenharia de Computadores e Informática (PECI)

- Demonstrations.
- Writing documentation and reports.

REFERENCES

Brás, J. (2022). *Virtual Assistant for Meeting Coordinators*. Dissertação de Mestrado em Engenharia de Computadores e Telemática, Universidade de Aveiro.

Barros, F., Valente, A. R., Teixeira, A., & Silva, S. (2022, November). Harnessing the Role of Speech Interaction in Smart Environments Towards Improved Adaptability and Health Monitoring. In *International Conference on Wireless Mobile Communication and Healthcare* (pp. 271-286). Cham: Springer Nature Switzerland.

NUMBER OF STUDENTS

4 or 5

RELATION WITH PROJECTS

This proposal is related to project VITALITY funded by FEDER through COMPETE 2030, under the project COMPETE2030-FEDER-00755100 (Refª FCT: 16385) and Casa Viva+, funded in its initial phase (2022-2023) by OLI and INOVADOMUS associates.